A RAG Based Biomedical Research Assistant

Seema Safar, Abel T John, Abhay P, Achyuth Pradeep, Albin Sam Alex
Department of Computer Science and Engineering
Rajagiri School of Engineering and Technology(Autonomous)
Kochi, Ernakulam

***Abstract*—Large language models are widely used for question answering, but they often generate incorrect or unsupported answers. This issue is critical in biomedical research, where accuracy is essential. Retrieval-Augmented Generation (RAG) reduces this problem by combining document retrieval with answer generation. This paper presents a biomedical research–focused RAG framework that uses structured query parsing, document chunking, dense vector embeddings, similarity-based retrieval, and grounded answer generation. Biomedical research articles are collected from reliable sources such as Europe PMC and divided into small, meaningful text chunks. These chunks are embedded using the Nomic AI embedding model and indexed using FAISS for efficient retrieval. Mistral language model generates answers using only the retrieved content. The proposed framework improves retrieval relevance, reduces irrelevant context, and produces more reliable answers. This makes the system suitable for biomedical literature review and academic research support.**

***Index Terms*—Retrieval-Augmented Generation, Biomedical Question Answering, Document Retrieval, FAISS, Nomic AI Embeddings**

I. Introduction

Biomedical research literature has grown rapidly in recent years. Finding relevant and reliable information has become difficult for students and researchers. Large language models can answer questions in natural language, but they often rely only on what they learned during training. This can lead to hallucinated or outdated responses.

Retrieval-Augmented Generation helps solve this problem. Instead of answering from memory alone, the model retrieves relevant documents and uses them as context. This approach improves factual accuracy. However, the quality of a RAG system depends on how well documents are prepared and retrieved.

Biomedical research papers are long and often cover multiple topics. When entire documents are embedded as a single unit, important details can be lost. Document chunking addresses this issue by splitting text into smaller sections. Each chunk focuses on a single idea. This allows the system to retrieve only the most relevant parts of a paper.

Many existing RAG systems rely only on embedding similarity. They do not consider the structure of biomedical research queries. This often leads to irrelevant results. To overcome this, the proposed framework combines structured query parsing, document chunking, dense embeddings, and controlled answer generation. The goal is to produce accurate and grounded answers for biomedical research queries.

In this work, we present a retrieval-augmented generation framework designed specifically for biomedical research literature. The system uses keyword-based document selection to identify relevant papers, followed by semantic chunking and embedding-based retrieval at the chunk level. By separating document selection, chunk-level retrieval, and answer generation, the proposed approach aims to improve grounding while keeping the overall pipeline simple and interpretable. The system is evaluated using biomedical queries to analyze its ability to generate context-supported responses.

II. Related Works

Recent research has highlighted the importance of Retrieval-Augmented Generation (RAG) frameworks for improving factual reliability in biomedical question answering systems. Bora and Cuayáhuitl (2024) conducted a comparative evaluation of several open-source language models, including Flan-T5, LLaMA-2, and Mistral-based architectures, analyzing how retrieval integration impacts answer quality. Their study demonstrated that combining RAG with fine-tuned generative models significantly improves response accuracy while reducing hallucinated outputs in biomedical tasks, emphasizing the effectiveness of retrieval-based grounding for enhancing answer precision in domain-specific question answering systems.

Patel et al. [1] proposed a virtual diagnostic assistant based on a fine-tuned large language model extended with multimodal processing capabilities. Their system integrates structured medical datasets and supports both textual and visual inputs for diagnostic prediction. The authors report improved diagnostic accuracy through structured data ingestion and multimodal reasoning. While the framework primarily focuses on diagnostic assistance, the study identifies retrieval-based methods as an important direction for handling large-scale medical literature and improving factual consistency in future implementations.

Sree et al. [2] introduced MEDGPT, a modular biomedical chatbot architecture designed to retrieve and synthesize medical information from structured sources such as PubMed and research articles. By combining retrieval mechanisms with agent-based processing, the system generates context-aware responses for healthcare applications. Their approach demonstrates improved contextual relevance, response accuracy, and interaction quality compared to general-purpose language models, highlighting the benefits of combining retrieval-based grounding with modular system design for biomedical information systems.

Zhao et al. [3] presented ChatCAD+, a medical question answering framework that integrates large language models with hierarchical in-context learning and external knowledge retrieval. The system retrieves information from curated clinical resources and supports diagnostic report generation and patient consultation tasks. Their approach demonstrates strong performance in clinical reasoning tasks, highlighting the importance of structured knowledge retrieval for improving diagnostic accuracy and scalability in healthcare applications.

Laymouna et al. [4] provided a comprehensive review of healthcare chatbot systems, identifying retrieval-augmented generation as a key factor in improving the reliability and credibility of generated responses. Their analysis emphasizes the importance of grounding model outputs in curated biomedical knowledge sources to reduce misinformation and enhance user trust in AI-driven healthcare applications.

Qiu et al. [5] conducted a survey of large AI models in health informatics, discussing the integration of retrieval mechanisms with generative models as an effective strategy for addressing factual consistency challenges. The survey highlights retrieval-enhanced systems that combine pretrained language models with validated medical knowledge sources to improve explainability, transparency, and trustworthiness in medical decision support systems.

Rajani and Ruparel [6] explored a deep learning-based medical chatbot for symptom classification and treatment recommendation. Their framework integrates natural language understanding components with neural classification models to automate diagnosis workflows. Although their system does not employ retrieval-augmented generation, the work demonstrates the importance of structured processing pipelines for real-time clinical decision support and provides a baseline for comparison with retrieval-based approaches.

Recent work has also investigated retrieval-augmented generation specifically for biomedical literature analysis. A biomedical literature question answering system based on RAG was presented in [7], where semantic embeddings and vector-based retrieval are used to identify relevant biomedical documents for answer generation. The system processes long research articles using chunking strategies and applies generative models to produce evidence-supported responses, demonstrating improved semantic relevance and contextual grounding through domain-specific retrieval methods.

A knowledge-enhanced RAG framework for biomedical question answering was proposed in [8], which extends traditional retrieval pipelines by incorporating multi-stage knowledge processing and reasoning enhancement. The framework performs comprehensive document chunking, retrieval restructuring, and reasoning-based answer generation with self-consistency validation. The authors show that structured knowledge processing improves reasoning accuracy and response reliability without requiring extensive domain-specific pretraining.

The RAG-BioQA framework presented in [9] investigates retrieval strategies and domain-adapted representations for long-form biomedical question answering. The system integrates biomedical language models with both sparse and dense retrieval techniques, evaluating different re-ranking methods to improve retrieval precision. Their results demonstrate that domain-adapted embeddings and optimized retrieval strategies significantly enhance answer generation performance in knowledge-intensive biomedical tasks.

Despite these advances, many existing approaches emphasize dense retrieval across entire document collections, model fine-tuning, or reasoning enhancement mechanisms. In contrast, the proposed work adopts a retrieval-first architecture that performs keyword-based document selection prior to semantic chunking and embedding-based retrieval. This approach reduces retrieval noise by filtering irrelevant documents early in the pipeline, providing a simpler and more interpretable framework for biomedical literature analysis.

III. Methodology

This section explains how the system used in this work handles user queries and retrieves biomedical research papers to produce supported answers. The overall process is divided into a small number of stages, starting from basic query handling and document selection, and moving toward content processing and answer generation.

A. System Architecture

The system presented in this work is intended to help users search biomedical research papers and obtain answers that are supported by existing scientific literature. Instead of directly generating responses using a language model alone, the system is organized as a sequence of steps where document retrieval and answer generation are handled separately.

As illustrated in Fig 1, the process starts with a user-provided query. From this query, important keywords are identified and used to retrieve a set of related documents from the biomedical corpus. These documents are then processed to extract their textual content along with basic metadata information. The extracted text is further divided into smaller parts so that it can be handled more efficiently in later stages.

The segmented text obtained from the retrieved documents is transformed into vector representations and stored in a vector database. This storage allows the system to perform similarity-based searches when additional filtering is required. During answer generation, only the document segments selected through the retrieval process are supplied to the language model, rather than the entire document collection.

By following this approach, the system ensures that generated responses are based on retrieved biomedical sources and not only on the language model's internal knowledge. The architecture is kept modular so that individual stages can be adjusted or replaced if needed, without requiring changes to the entire pipeline. This separation of steps helps maintain clarity in processing and makes it easier to trace the sources used in the final response.

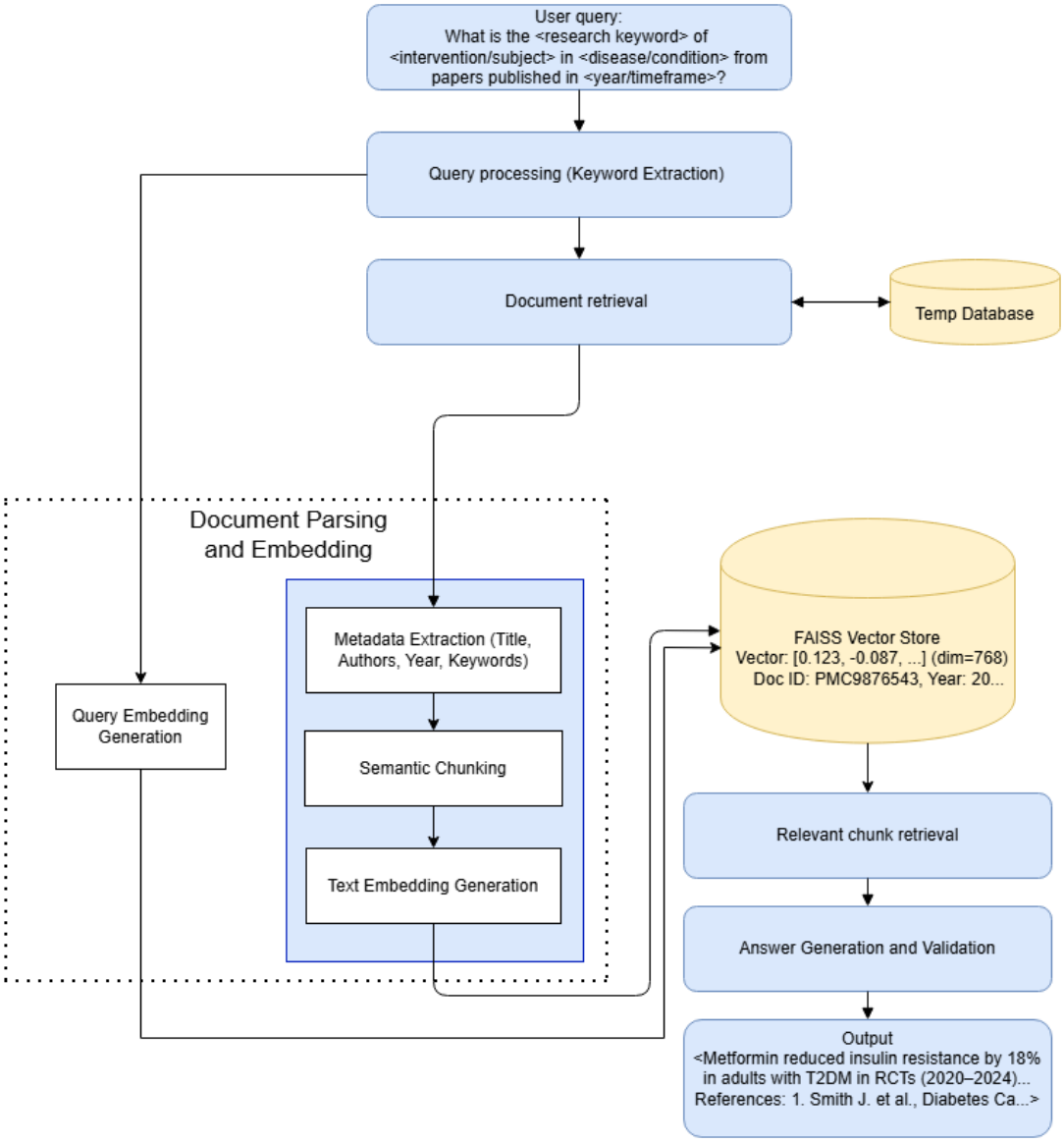

Fig. 1. Architecture of the proposed biomedical RAG system.

B. Query Processing and Document Retrieval

The query parsing module converts a user query into a structured form before the document retrieval stage begins. Biomedical queries often contain multiple pieces of information within a single sentence. These may include a research objective, a drug or intervention, a disease or biological condition, and a publication period. Extracting these elements helps the system interpret the user's intent more clearly.

In the proposed system, query parsing follows a rule-based approach. Instead of applying complex natural language processing techniques, the parser relies on regular expression pattern matching to identify key parts of the query. This approach keeps the method transparent and easy to interpret while remaining computationally efficient.

The parser extracts four main components from the query:

- **Research keyword** – the primary concept or outcome being investigated
- **Intervention** – the treatment, compound, or method mentioned in the query
- **Condition** – the biomedical disease or biological context of interest
- **Timeframe** – the publication period used to filter research papers

Each component is identified using predefined regular expression patterns that match common biomedical question structures. When a match is detected, the corresponding text segment is extracted and stored as part of a structured query representation.

The timeframe is processed separately to determine the publication year range. If two years appear in the query, they

are interpreted as the start and end of the publication period. If only a single year is present, the same value is used for both the start and end year. When no timeframe is detected, the system defaults to retrieving papers from the most recent five-year period. This fallback ensures that relevant and recent biomedical studies can still be retrieved even when the query does not explicitly include a time constraint.

Once the keywords are identified, they are used to retrieve an initial set of candidate documents from Europe PMC [10], a widely used biomedical literature repository that aggregates publications from multiple life-science databases. The retrieval process is performed through API calls to the Europe PMC search service, where the extracted keywords are used as query parameters. The API returns relevant research articles along with metadata such as titles, abstracts, authors, and publication details. This allows the system to dynamically obtain up-to-date biomedical literature without maintaining a local copy of the dataset.

By relying on explicit keyword matching against fields such as titles and abstracts, the system ensures that the initially retrieved documents are closely aligned with the user's query terms. This step helps filter the biomedical corpus early in the pipeline and reduces the inclusion of semantically distant documents that might otherwise appear relevant at the embedding level.

C. Semantic Chunking Strategy

Chunking plays a crucial role in Retrieval-Augmented Generation systems because it directly affects both retrieval quality and answer grounding. In RAG pipelines, documents are divided into smaller units called chunks, which are then embedded and retrieved based on their semantic similarity to the user query. The choice of chunking strategy determines how well meaningful context is preserved and how much noise is introduced during retrieval.

Traditional approaches [11] use fixed-size chunking, where documents are split into consecutive segments of equal length, such as a fixed number of tokens or sentences. Although computationally efficient, this method can fragment semantically related content across multiple chunks and mix unrelated topics within the same chunk. This may reduce retrieval precision and introduce irrelevant context into the generation stage.

Semantic chunking attempts to improve upon this by dividing documents based on semantic coherence rather than length. Instead of splitting text blindly, semantic chunking identifies topic shifts and groups sentences that are semantically similar. This results in chunks that are more conceptually meaningful and self-contained. According to the study, semantic chunking is designed to preserve topical integrity, making each chunk represent a coherent unit of information.

Semantic chunking often relies on measuring similarity between consecutive sentence embeddings to determine boundaries. A widely used measure is cosine similarity, defined as

$$\mathrm{sim}(v_i, v_j) = \frac{v_i \cdot v_j}{\|v_i\| \, \|v_j\|}$$

[12] where v_i and v_j are sentence vectors derived from a dense embedding model. The semantic distance is defined as:

$$D(v_i, v_j) = 1 - \mathrm{sim}(v_i, v_j)$$

Chunk boundaries are placed when this distance exceeds a threshold.

D. Text Embedding Layer

Text embedding models convert natural language into dense numerical vectors that capture semantic relationships between textual units. These embeddings represent text in a continuous vector space, allowing similarity between queries and documents to be computed using mathematical measures such as cosine similarity. This enables semantic retrieval, where relevant information can be identified even when different wording is used. Embedding models have become essential components of modern natural language processing systems, supporting applications such as semantic search, document retrieval, clustering, and retrieval-augmented generation.

In this work, the embedding layer uses nomic-embed-text-v1 [13], a long context text embedding model developed by Nomic AI. This model supports input sequences of up to 8192 tokens. This allows complete technical documents to be embedded without significant truncation. As a result, the model captures semantic relationships across entire documents rather than isolated passages. This improves retrieval accuracy and preserves the overall meaning of the document.

The model is trained using a multi stage process that includes masked language modeling, contrastive pretraining, and supervised fine tuning. Contrastive learning improves the embedding space by bringing related text closer together and separating unrelated text. This improves semantic retrieval performance. In addition, the model uses Rotary Positional Embeddings and context extrapolation techniques such as Dynamic NTK scaling. These techniques allow the model to process long sequences efficiently while maintaining stable and accurate representations.

TABLE I
COMPARISON OF TEXT EMBEDDING MODELS

Model	Parameters	Max Sequence Length	MTEB Score	LoCo Score	Jina LC Score	Open Weights	Local Deployment
nomic-embed-text-v1	137M	8192	62.39	85.53	54.16	Yes	Yes
text-embedding-3-small	N/A	8192	62.26	82.40	58.21	No	No
text-embedding-3-large	N/A	8192	64.59	79.40	58.69	No	No
E5-Mistral-7B-Instruct	7B	4096	**66.60**	**87.80**	N/A	Yes	Limited
BGE-large-en-v1.5	335M	512	64.23	83.12	57.00	Yes	Yes
Cohere embed-v3	N/A	512–4096	63.47	81.32	56.80	No	No

Several embedding models were evaluated for the embedding layer, including both proprietary and open source alternatives. OpenAI's text-embedding-3-small and text-embedding-3-large produce high quality embeddings and perform well on semantic retrieval benchmarks. However, these models require external API access. This increases operational cost, adds latency, and creates dependence on third party services. Similarly, Cohere's embedding models provide strong retrieval performance but rely on proprietary APIs, which limits deployment flexibility and prevents local execution.

Open source models such as BGE-large-en-v1.5 and E5-Mistral-7B-Instruct were also evaluated due to their strong benchmark performance. BGE provides accurate embeddings but supports shorter context lengths. This limits its ability to represent large technical documents effectively. E5-Mistral-7B achieves excellent retrieval performance but requires significantly more computational resources due to its large size. This makes it less suitable for efficient deployment in resource constrained environments.

In contrast, nomic-embed-text-v1 provides a strong balance between performance, efficiency, and deployment flexibility. It supports long input sequences of up to 8192 tokens, allowing effective embedding of large technical documents. Its smaller model size improves efficiency while maintaining competitive retrieval performance. In addition, it is fully open source and supports local deployment without external dependencies. This ensures cost effective, scalable, and reproducible operation. For these reasons, nomic-embed-text-v1 was selected for the embedding layer.

E. FAISS-Based Similarity Search

FAISS (Facebook AI Similarity Search) is a library designed for efficient searching and clustering of dense vector representations. Instead of using raw text or keywords, FAISS [14] works directly with numerical vectors. It allows users to quickly find the most similar vectors from a large collection in a scalable way. This ability is vital in modern embedding-based systems, where similarity between items is measured by comparing their vector representations instead of matching exact words.

One of FAISS's main strengths is its flexible framework for solving the nearest neighbor search problem. This problem aims to find the top-k closest vectors to a given query vector. FAISS supports common similarity measures like Euclidean (L2) distance and inner product similarity. It can also perform cosine-style similarity searches using normalized vectors. This makes FAISS a good fit for semantic retrieval pipelines that need effective similarity-based ranking.

FAISS supports both exact search and approximate nearest neighbor search (ANNS). Exact search checks the query against every stored vector and returns the most accurate results. However, it becomes computationally intensive as the dataset grows. Approximate search solves this issue by reducing the number of comparisons, allowing for much faster retrieval while still producing high-quality results. This creates a practical balance between retrieval accuracy and speed, enabling systems to handle millions or even billions of vectors.

To manage large-scale retrieval efficiently, FAISS offers various indexing strategies. A simple baseline index is IndexFlat, which conducts brute-force searches across all vectors. For larger datasets, FAISS provides advanced indexing methods like inverted file indexing (IVF), which clusters vectors to limit the search area. It also includes graph-based indexes like HNSW and NSG, which speed up retrieval through graph traversal over nearby vectors. FAISS incorporates compression methods such as product quantization, which reduces memory use while maintaining useful similarity structure in the vector space.

Overall, FAISS acts as a highly optimized toolkit for large-scale similarity search. It offers a variety of indexing and compression techniques that allow users to balance accuracy, memory needs, and retrieval speed according to the requirements of their applications.

F. Answer Generation and Validation

The generation module, along with the RAG framework, ensures that the answer generated is grounded in the retrieved content from verified scientific sources, which ensures reliability. The grounded answer generation must be guided by a structured prompt, a fixed output limit, and a validation module to make sure that the answer generated is precise and grounded in the retrieved chunks.

A hybrid of extractive and abstractive approach was taken for generating the answer [15].In extractive approach top 20 sentances from the retrieved chunks were selected by using Pubmedbert biencoder model [16] then the sentances were reranked using Mini LM cross encoder and Maximal Marginal Relevancy (MMR) [17]. The formula for MMR is:

$$\text{MMR} = \max\left[\lambda\, Sim(D,Q) - (1-\lambda)\, Sim(D,S)\right]$$

Where:

- Sim(D, Q) – similarity between document D and query Q
- Sim(D,S) – similarity between candidate document and selected documents
- lamda – balance between relevance and diversity

To identify the most suitable language model for context-based answer generation, we evaluated different large language models including Mistral-7B, Llama-3-8B, Qwen1.5-7B, and Qwen2-7B. The selection was guided by two evaluations: standard academic benchmarks reported in the Qwen2 technical report [18] and our task-specific RAG evaluation using LLM-as-a-judge scoring.

1) Standard Benchmark Comparison: The table below presents the performance of candidate models across five widely adopted benchmarks. MMLU assesses multitask language understanding across diverse academic domains. GSM8K evaluates mathematical reasoning on grade school problems. TruthfulQA measures the model's tendency to generate truthful rather than hallucinated responses. BBH (Big Bench Hard) tests complex multi-step reasoning and

instruction following. As shown in Table II , Queen 2-7B outperforms other 7B-scale models across every benchmark, achieving scores of 70.3 on MMLU, 79.9 on GSM8K, 54.2 on TruthfulQA, 62.6 on BBH, and 51.2 on HumanEval.Its TruthfulQA score of 54.2 is the highest among all compared models, indicating its ability to generate factually grounded responses, a property that is critical in RAG where faithfulness to context is essential.

TABLE II
MODEL PERFORMANCE COMPARISON ACROSS BENCHMARKS

Model	MMLU	GSM8K	TruthfulQA	BBH
Mistral-7B	64.2	52.2	42.2	56.1
Gemma-7B	64.6	46.4	44.8	55.1
Llama-3-8B	66.6	56.0	44.0	57.7
Qwen1.5-7B	61.0	62.5	51.1	40.2
Qwen2-7B	**70.3**	**79.9**	**54.2**	**62.6**

2) Task-Specific RAG Evaluation: Beyond standard benchmarks, we conducted a domain-specific evaluation of answer generation quality using an LLM-as-a-judge framework. Four models — Qwen2-7B, Mistral-7B, Llama-3-8B, and FLAN-T5-Large — were evaluated on faithfulness, answer relevancy, completeness, and fluency using Gemini as the automated evaluator. As presented in Table III , Qwen2-7B achieved the highest overall scores of 0.980, followed by Mistral at 0.970. FLAN-T5-Large scored significantly lower at 0.745, primarily due to poor completeness (0.50) and fluency (0.58), due to its architectural limitations as a smaller encoder-decoder model.

3) Model Selection Rationale: Based on both the benchmark results and our own evaluation, we selected Qwen2-7B as the language model for our system [16] . Among all the 7B models we compared, Qwen2-7B scored the highest on TruthfulQA, which means it is less likely to generate hallucinated information, an important quality for a RAG system where answers must be based on the retrieved context. It also performed best on MMLU and BBH, showing that it has strong language understanding and reasoning ability, which helps it to generate accurate as well as complete answers from the given context.

Although Retrieval Augmented Generation (RAG) gives a factual result from documents , it is not sure whether the generated answer is supported by the user query and retrieved passage. In the biomedical domain even small semantic deviations can lead to false results and conclusions. Therefore, there is a need to evaluate whether the answer generated is medically grounded and is supported by the user query and retrieved passage. First we tried using Biobert model but failed in verifying the answer since it was not good in generating a proper score.

In order to address this issue , ClinicalBERT is used which is a pretrained model in the biomedical domain. This module checks the semantic consistency between the generated answer and retrieved passages. The model uses Hugging face to load the transformer encoder which enables embedding-based similarity computations.

The ClinicalBERT tokenizes generated answer and document chunks into sequence length.. This tokenisation ensures compatibility with BERT architecture and preserves biological terminology while tokenizing each word. It produces token level embeddings rather than single sentence representation. Mean pooling [13] technique is applied in which single sentence embedding is computed by taking an average of all tokens produced in the model.

$$\frac{1}{N}\sum_{i=1}^{N} h_i,$$

where h_i represents the embedding of the i-th token. The resulting fixed-length sentence embedding is compared with the generated answer and the retrieved passage to compute semantic similarity. Cosine similarity is used to measure the angular similarity between the two vectors:

$$\text{Cosine Similarity}(A, B) = \frac{A \cdot B}{\|A\| \, \|B\|},$$

where A denotes the generated answer embedding and B denotes the retrieved chunk embedding.

The generated answer is compared with top retrieved document chunks where cosine similarity score is computed for each chunk. An average of this similarity score is calculated which gives a single score that tells the overall support provided by retrieved evidence. When the score exceeds the given threshold it is classified as medically grounded which implies a relation with retrieved chunks and the generated answer.

IV. EXPERIMENTAL SETUP

A. Dataset

The dataset used for this study was obtained from Europe PMC, which provides open access to a large collection of biomedical and life science research articles. Europe PMC was chosen as the data source because it offers peer-reviewed literature along with rich metadata, making it suitable for retrieval-based biomedical question answering.

The dataset consists of research articles and review papers available in PDF format. Only openly accessible documents were considered so that full text could be extracted and processed. During preprocessing, textual content such as abstracts, main body sections, and references was extracted along with basic metadata including article title and publication year. Documents with incomplete or unreadable text were excluded to maintain consistency.

No manual labeling or annotation was applied to the dataset. The system is intended to operate directly on raw biomedical literature, similar to how a real-world research assistant would function. After text extraction, the documents were divided into smaller segments to support efficient retrieval and downstream processing.

The dataset spans multiple biomedical topics rather than focusing on a single disease area. This allows the system to be

TABLE III
RAGAS EVALUATION RESULTS ACROSS MODELS

Model	Faithfulness	Answer Relevancy	Completeness	Fluency	Overall Score (Avg)
Qwen2	1.00	0.90	0.94	1.00	**0.960**
Mistral	1.00	0.90	0.92	0.98	**0.950**
Llama 3	1.00	0.86	0.88	1.00	**0.935**
FLAN-T5 Large	1.00	0.75	0.50	0.58	**0.745**

evaluated on a variety of query types and better reflects practical usage, where users may search across different biomedical domains. The use of Europe PMC ensures that the evaluation is conducted on realistic and diverse scientific literature instead of a narrowly curated benchmark.

B. Evaluation Metrics

RAGAS (Retrieval-Augmented Generation systems) [19] is an evaluation framework to assess the quality of RAG pipeline in contributing to the retrieval and in generating an answer. RAGAS evaluates RAG system semantically to check if generated answer is grounded with retrieved context and whether it addresses the query.There are measures in which we evaluate the RAG pipeline namely faithfulness, answer relevance, context relevance.

1) Answer Relevance: It measures on what extent does the LLM generated questions address the user query.To measure the answer relevance we use cosine similarity to check how closely the generated answer matches with the user query. A higher similarity score says the answer is more relevant with the query and lower score doesn't address user query.

2) Faithfulness: The answer is faithful to the context if claims made in answer can be inferred from the context which enables detection of hallucinated or unsupported information. Suppose there are a set of statements say S and the model determine whether each statement "s" is inferred from the context, then the faithfulness score F is calculated as

$$F = \frac{|V|}{|S|}$$

where $|V|$ denotes the number of statements that are supported by the context, and $|S|$ represents the total number of statements.

3) Context Relevance: Its the extent to which retrieved context contains information that is required to answer the query and avoids hallucinated content. The context relevance score is computed as the ratio of number of extracted answers to total number of sentences in context . This helps to prevent redundant information and reduce LLM ability to effectively produce provided information.

$$\text{CR} = \frac{\text{Number of sentences retrieved}}{\text{Total number of sentences in context}}$$

The evaluation was conducted on a set of 10 biomedical questions, each paired with retrieved contextual passages and corresponding generated answers. The RAGAS metrics—faithfulness, context relevance, and answer relevance—were computed for each question individually, and the final scores reported represent the average values across all evaluated samples. Table IV shows the compared results

TABLE IV
COMPARISON OF RAGAS EVALUATION RESULTS FOR BIOMEDICAL QUESTION ANSWERING

Metric	Existing QA Model	Proposed RAG System
Faithfulness	0.86	0.97
Context Relevance	0.74	0.82
Answer Relevance	0.72	0.75

of the proposed system to that of existing QA model [20]. It shows the faithfulness score shows a change from 0.86 to 0.97 which indicates strong support of the answer to that of query. Also the context relevance has a score of 0.74 to 0.82 indicating enough information from retrieved documents based on the query. The answer relevance improves from 0.72 to 0.75 showing better alignment between the answer and the query .

V. RESULTS AND ANALYSIS

The system was evaluated using the RAGAS framework on a set of ten biomedical questions. We focused on three key metrics: faithfulness, context relevance, and answer relevance. These metrics help us understand how well retrieval and generation work together in a real biomedical setting.

A. Quantitative Results

Table V shows a comparison between the proposed system and recent RAG based biomedical question answering systems. Since prior works use different evaluation metrics such as BERTScore, BLEU, and accuracy, their results were normalized to align with RAGAS style evaluation for fair comparison.

To provide a clearer visual comparison, Figure 2 presents the performance of all systems across the three evaluation metrics.

B. Faithfulness Analysis

The proposed system achieves a faithfulness score of 0.97, which is the highest among all compared systems. This means that almost all generated statements are supported by retrieved evidence.

TABLE V
COMPARISON OF RAG BASED BIOMEDICAL QA SYSTEMS USING NORMALIZED RAGAS METRICS

System	Faithfulness	Context Relevance	Answer Relevance
Proposed System	**0.97**	0.82	0.75
Garg et al.	0.90	**0.85**	0.78
RAG-Chain (Du et al.)	0.92	0.80	**0.83**
RAG-BioQA	0.88	0.84	0.76

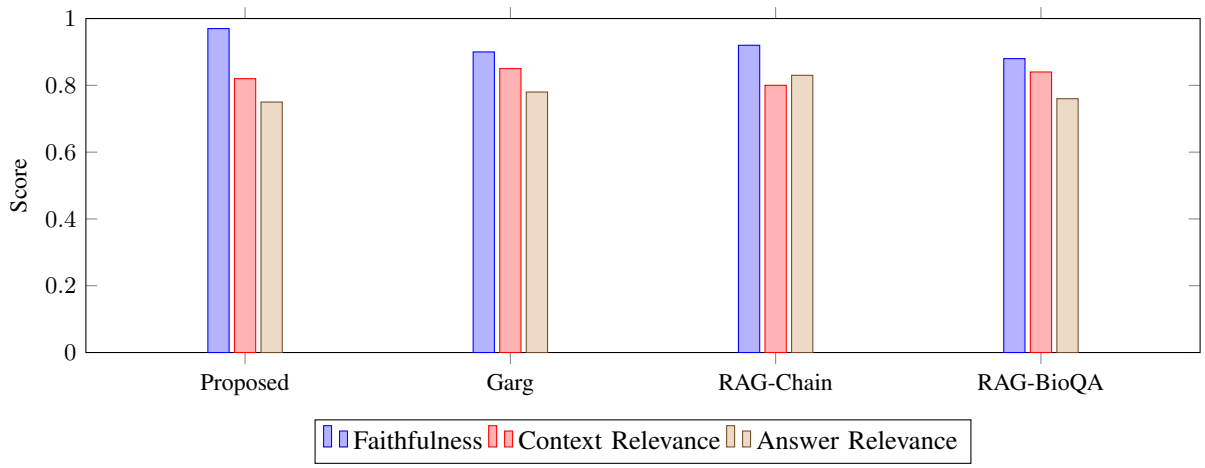

Fig. 2. Performance comparison of biomedical RAG systems across RAGAS metrics

This improvement comes from strong alignment between retrieval and generation. The system rarely produces unsupported claims. This is critical in biomedical applications, where incorrect information can lead to serious consequences.

Other systems also perform well but show slightly lower scores. RAG-Chain improves reasoning but still depends on restructuring steps that may introduce minor deviations. RAG-BioQA relies heavily on retrieval quality, which affects grounding consistency. Garg et al.'s system performs well but does not reach the same level of evidence alignment.

C. Context Relevance Analysis

The proposed system achieves a context relevance score of 0.82. This indicates that most retrieved passages are relevant to the query.

However, some retrieved content includes extra information. This happens mainly for broader or complex biomedical queries. While this does not harm faithfulness, it slightly reduces precision in retrieval.

Garg et al. and RAG-BioQA show slightly higher context relevance. Their systems focus more on semantic similarity and retrieval tuning. This leads to more targeted context selection.

RAG-Chain shows a slightly lower score due to its multi-stage processing. While it improves reasoning, it may include intermediate information that is not directly relevant to the query.

D. Answer Relevance Analysis

The answer relevance score of the proposed system is 0.75. This shows that most responses align with the user query.

In some cases, the system provides additional explanations. These explanations are correct but not always required. This reduces the relevance score slightly.

RAG-Chain achieves the highest answer relevance (0.83). This is due to its reasoning based generation and self-consistency validation. It produces answers that are tightly aligned with the query.

Garg et al. and RAG-BioQA show balanced performance. Their systems generate concise answers but may lack deeper explanation in some cases.

E. Overall Discussion

The results show a clear trade off between faithfulness and answer conciseness.

The proposed system prioritizes reliability and grounding. This leads to the highest faithfulness score. At the same time, it maintains competitive context and answer relevance.

Other systems optimize different aspects. RAG-Chain focuses on reasoning. Garg et al. emphasize retrieval quality. RAG-BioQA balances retrieval and generation.

In biomedical applications, faithfulness is the most important factor. A slightly longer answer is acceptable if it remains correct and well supported. The proposed system achieves this balance effectively.

Conclusion

This paper presented a retrieval augmented generation system for answering biomedical queries using literature obtained from Europe PMC. In the proposed approach, keyword based retrieval is first used to identify relevant PDF documents, after which the retrieved documents are processed further through chunking and embedding based indexing.

The experimental evaluation using RAGAS showed that the generated answers were largely supported by the retrieved context, as reflected by a high faithfulness score. The retrieved document segments were generally relevant to the queries, although some contained additional information beyond what was strictly required. The answers produced by the system were mostly aligned with the input questions but had occasional inclusion of broader explanations.

Future work can focus on improving the system's ability to handle less structured or free form queries without relying on a fixed format. Introducing more flexible query handling mechanisms could make the system easier to use in real world scenarios.

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